CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Memory Organization

Lecture No.- 3

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Recap of Previous Lecture

Topic Memory Address Decoder

Topic Main Memory: RAM, ROM

Topic RAM Chip

Topic ROM Chip

Topics to be Covered

Topic Multiple Chips in Single Memory System

Topic DRAM Refresh

Topic : Multiple Chips in Single Memory System

Total memory capacity = no. of chips * 1 chip capacity

[NAT]

#Q. How many 64 bytes RAM chips are needed to provide a memory capacity of 1Kbytes?

$$\frac{1KB}{64B} = \frac{2^{10}}{2^6} = 2^4 = \underline{\underline{16}} \text{ Ans.}$$

[NAT]

#Q. Total memory capacity is <u>8</u> Mbytes, if we use 16 chips of size 512Kbytes each?

$= \underline{16} * \underline{512K}B$

$= 2^{23} B$

$= 8MB$

Topic : Multiple Chips in Single Memory System

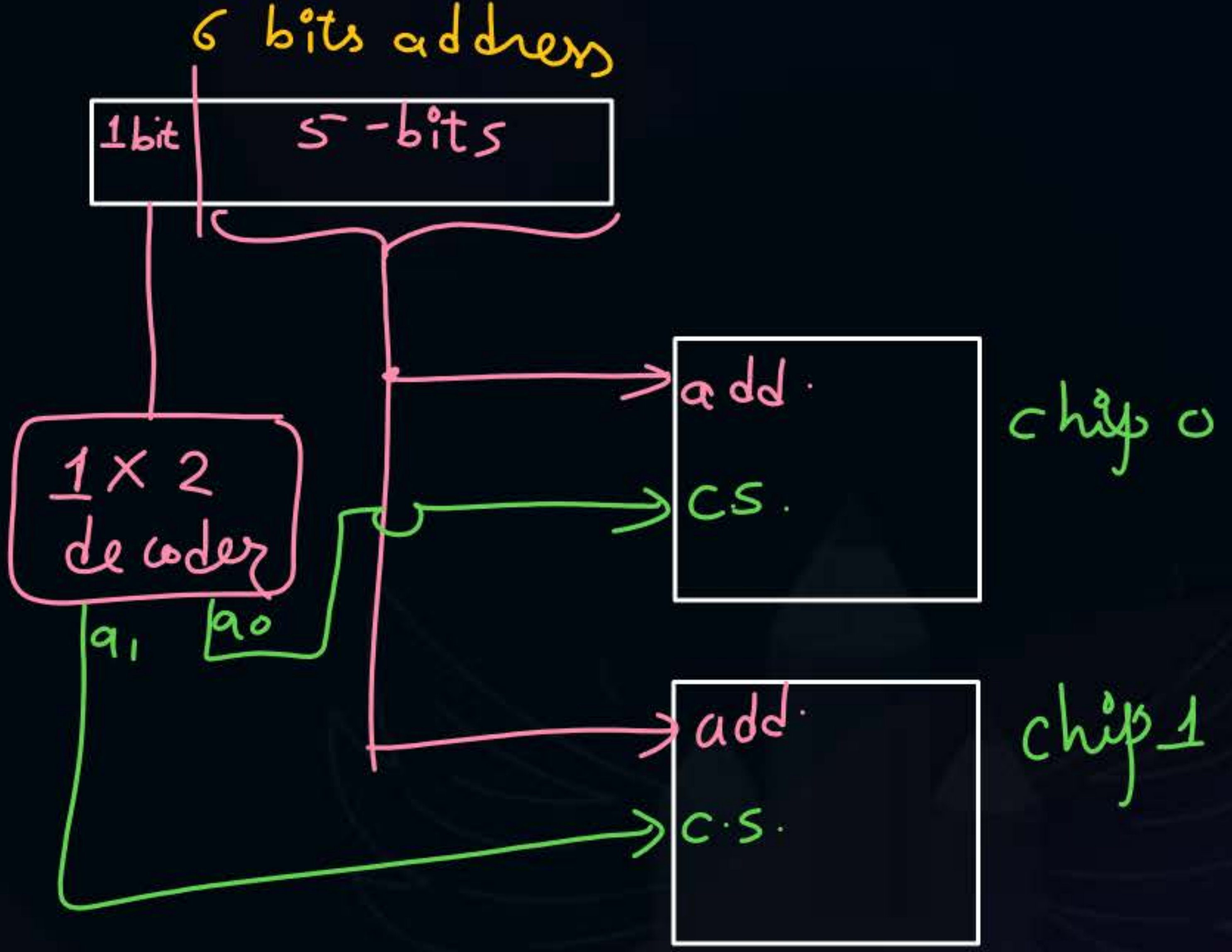
32 × 8 bits chip 0 ⇒ 5 bits

32 × 8 bits chip 1 ⇒ 5 bits

64 × 8 bits ⇒ address = 6 bits

6 bits CPU generates

0	000000	chip 0
⋮	⋮	
31	011111	
32	100000	chip 1
⋮	⋮	
63	111111	

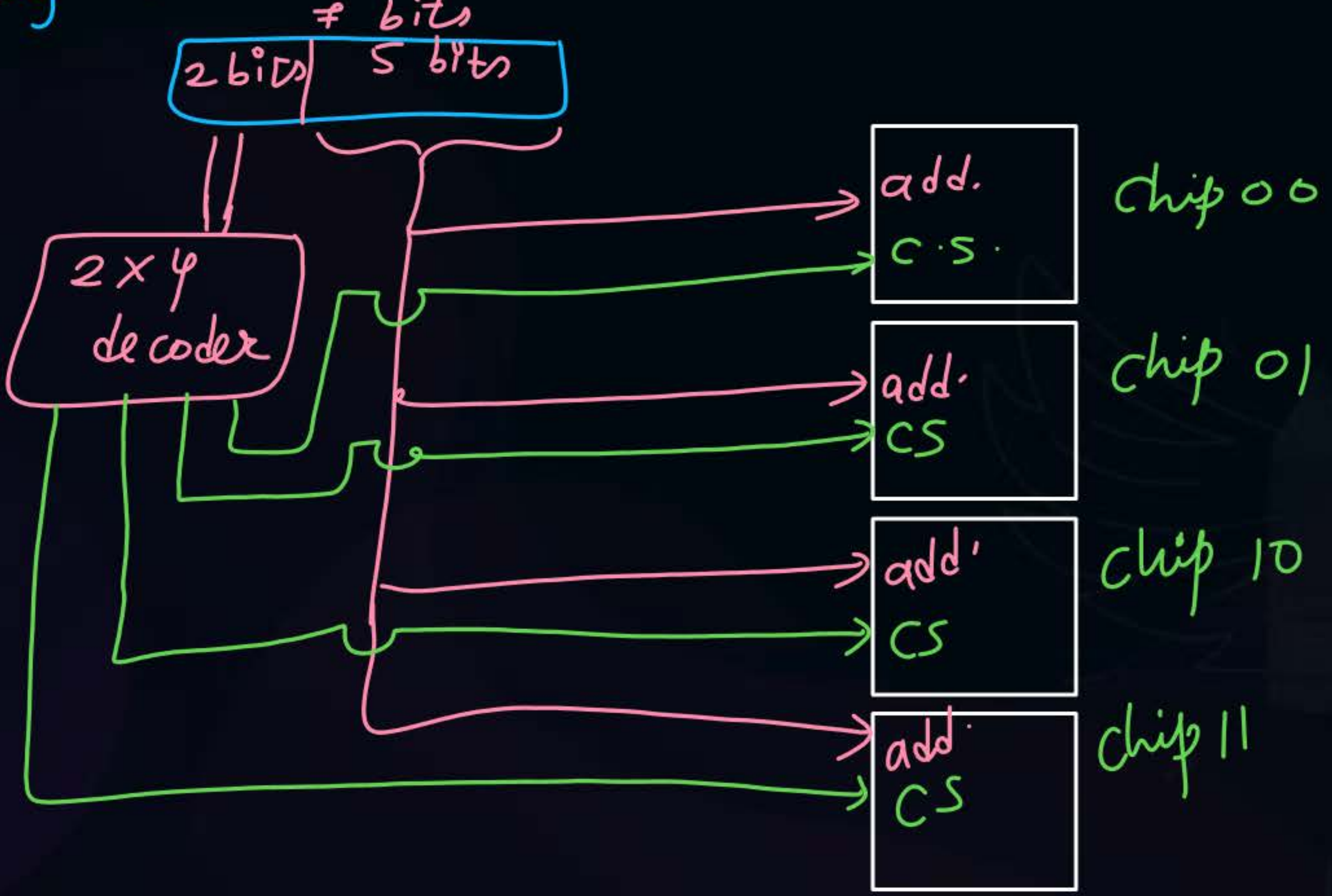


Topic : Multiple Chips in Single Memory System

32×8 bits $\Rightarrow$ 4 chips

$\Downarrow$

Total capacity $= 128 \times 8$ bits $\Rightarrow$ add. $= 7$ bits



vertical arrangement $\Rightarrow$ needed when no. of addresses only are more in total mem. system than single mem. chip.

[NAT]

$\rightarrow 2^7 \Rightarrow 7$ lines

#Q. (a) How many 128×8 bits RAM chips are needed to provide a memory capacity of 2048 bytes?

(b) How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips?

(c) How many lines must be decoded for chip select? Specify the size of decoder?

(a) $\frac{2048 \cancel{B}}{128 \times \cancel{8\,bits}} = \frac{2^{11}}{2^7} = 2^4 = 16$

(b) 11 lines

7 lines common

(c) 4 lines

4×16 decoder

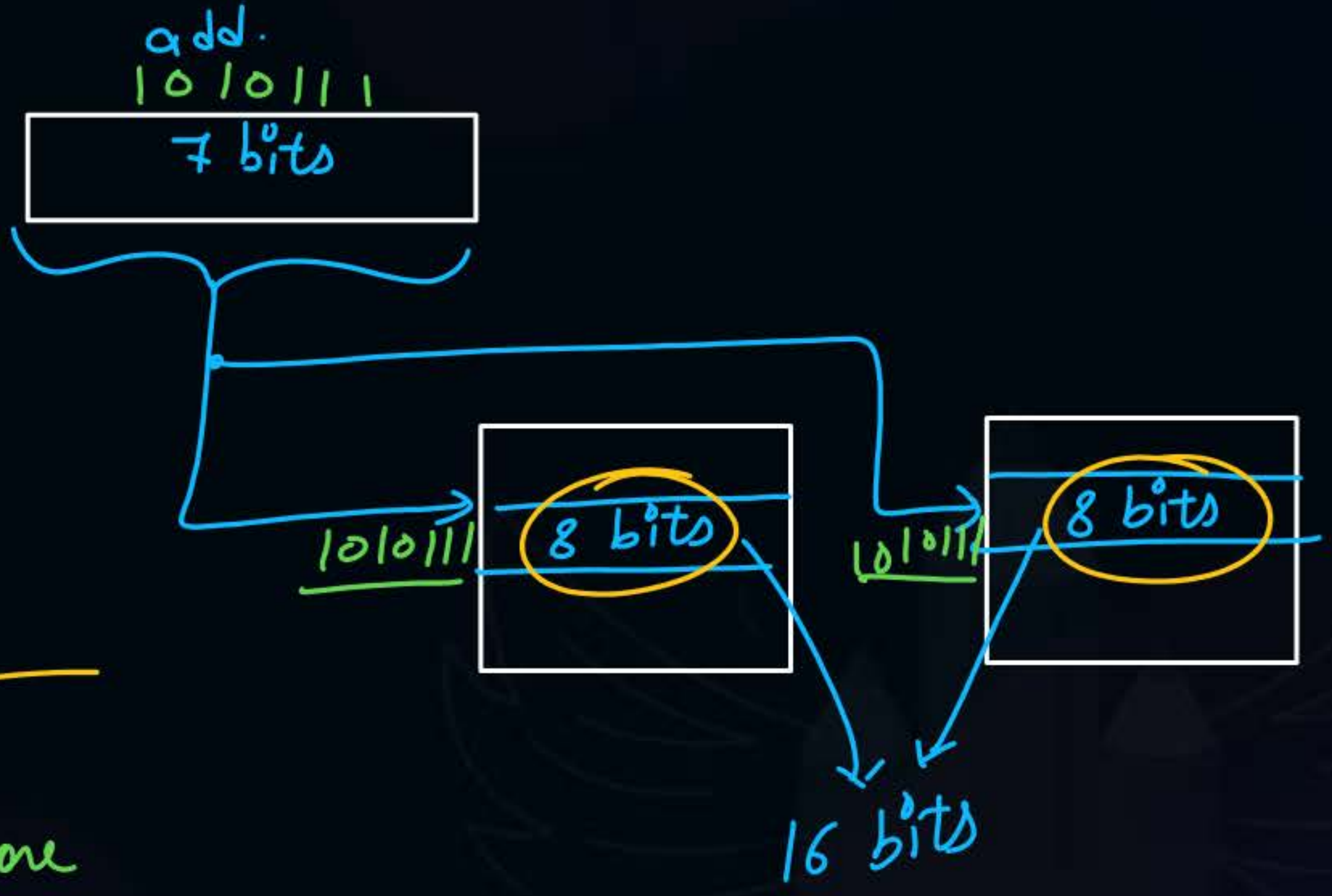
[NAT]

#Q. How many 128×8 bits RAM chips are needed to provide a memory capacity of 128 ×16 bits?

$$\frac{128 \times 16}{128 \times 8} = 2$$

CPU generates add. = 7 bits

chip can understand add. = 7 bits



Horizontal arrangement ⇒

when the mem. to build require more bits per address than bits per address in single chip.

[NAT]

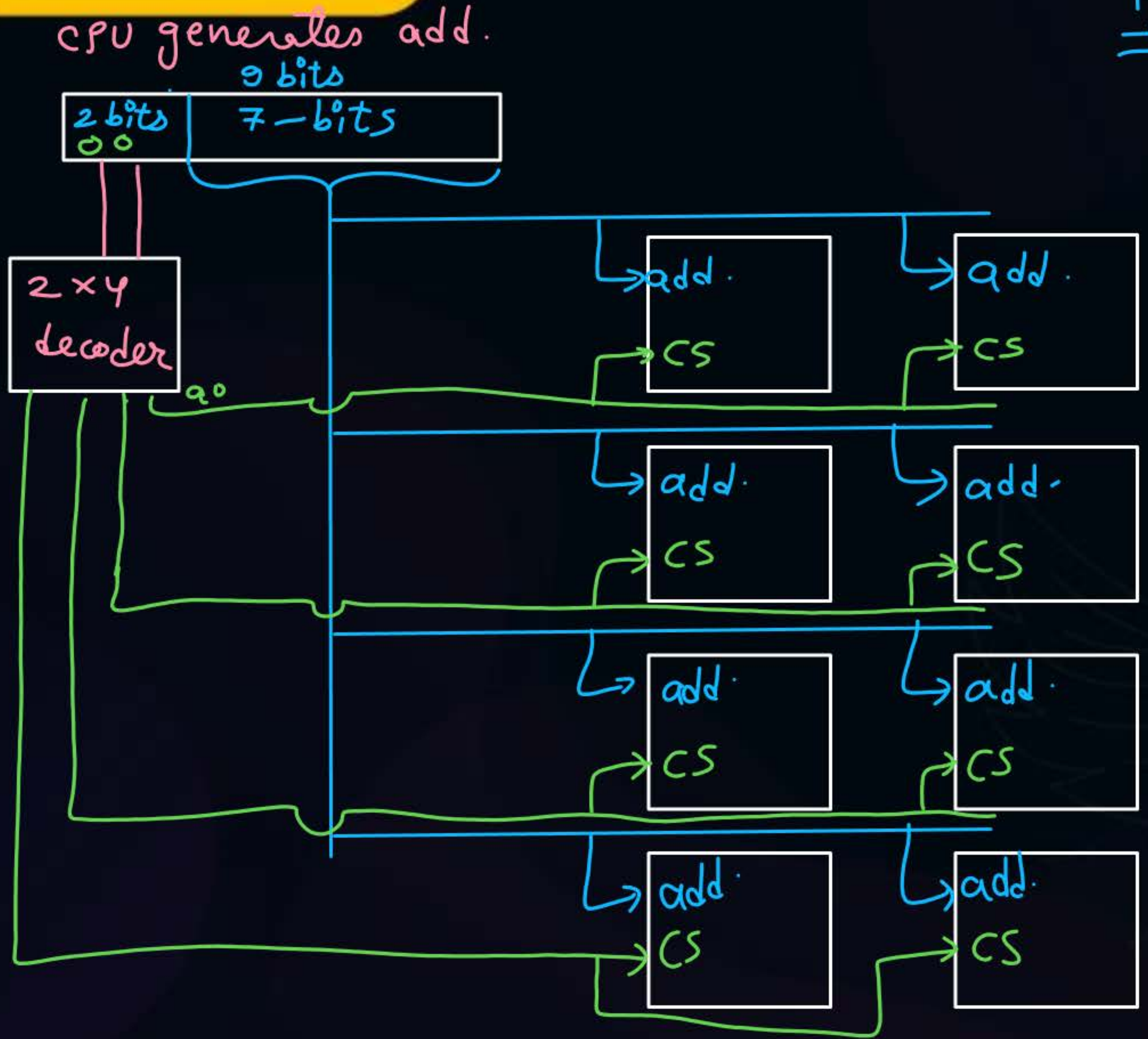
$\rightarrow 2^7 \Rightarrow 7$ bits

#Q. How many 128×8 bits RAM chips are needed to provide a memory capacity of 512 ×16 bits?

$$= \frac{\overset{4}{\cancel{512}} \times \overset{2}{\cancel{16}}}{\cancel{128} \times \cancel{8}} = 4 * 2 = 8 \text{ chips}$$

Topic : Solution

Hybrid



[MCQ]

GATE- 2009

#Q. How many 32K × 1 RAM chips are needed to provide a memory capacity of 256K bytes?

A 8

B 32

C 64 ✓

D 128

$$\frac{\cancel{256K}^{8}\ \cancel{\text{bytes}}^{8}}{\cancel{32K} \times \cancel{1}\ \text{bit}}$$

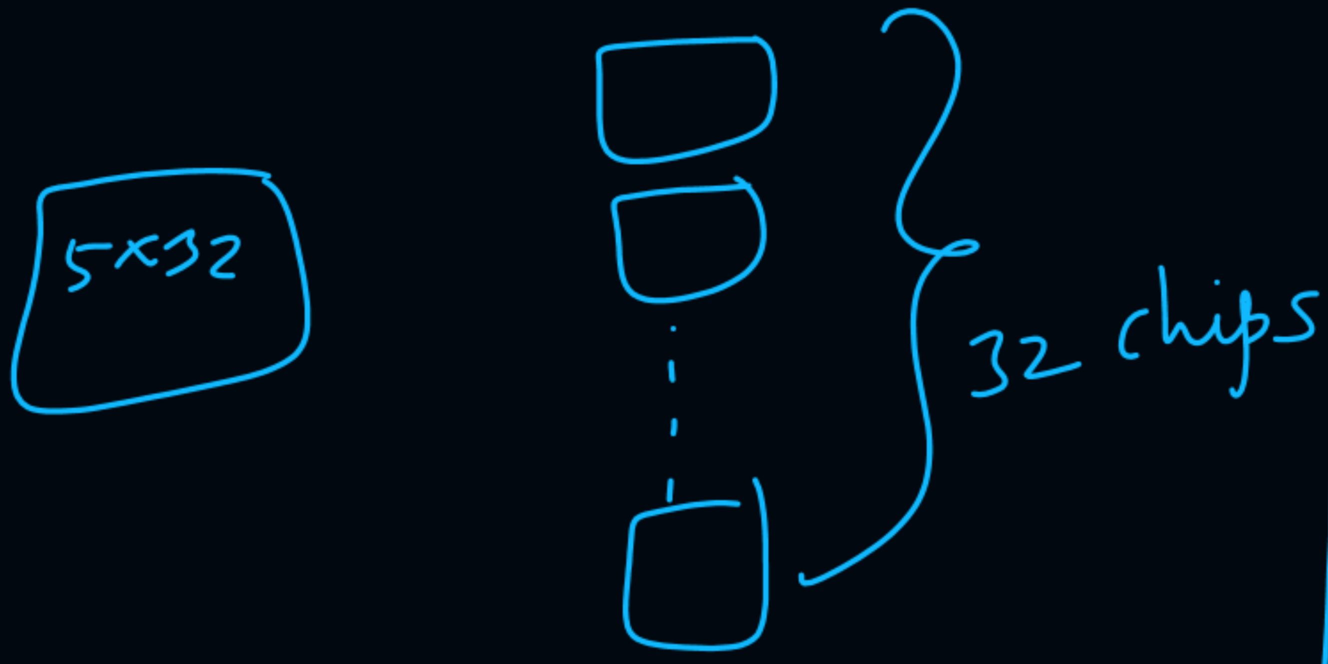
$= 64$ chips

default unit ⇒ bits

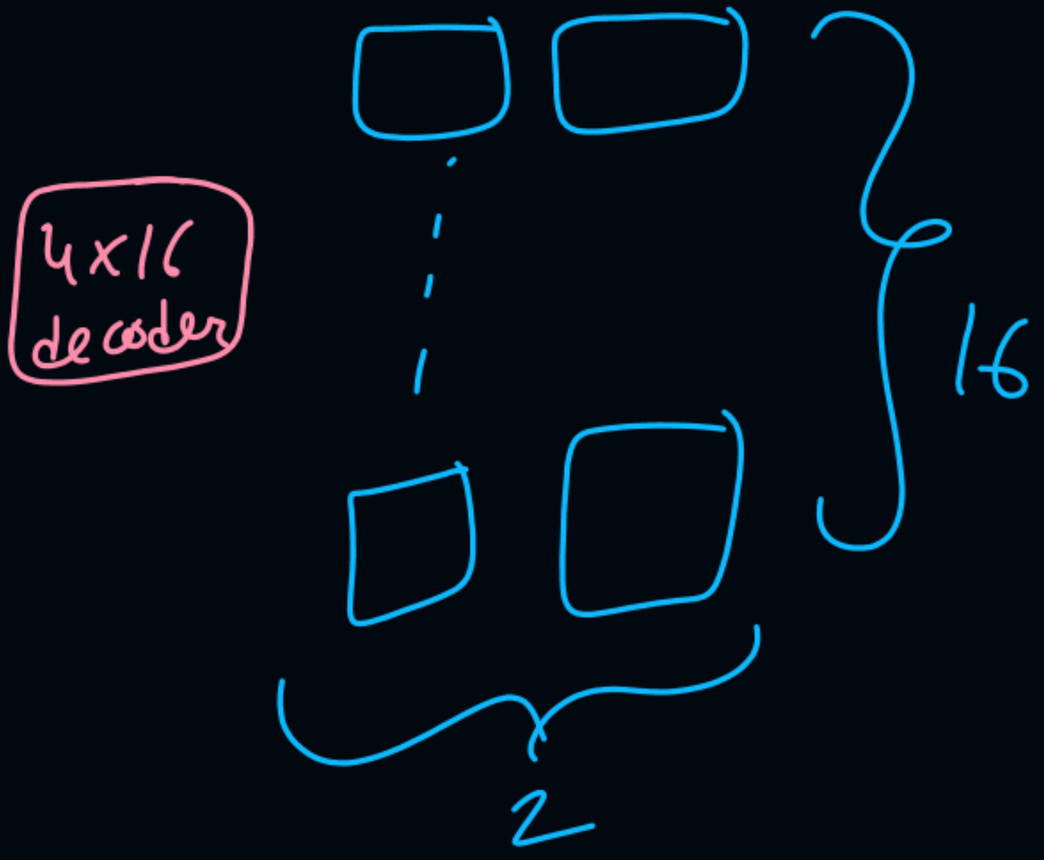
Ques) A mem. system has 32 chips total

5×32 decoder always needed?

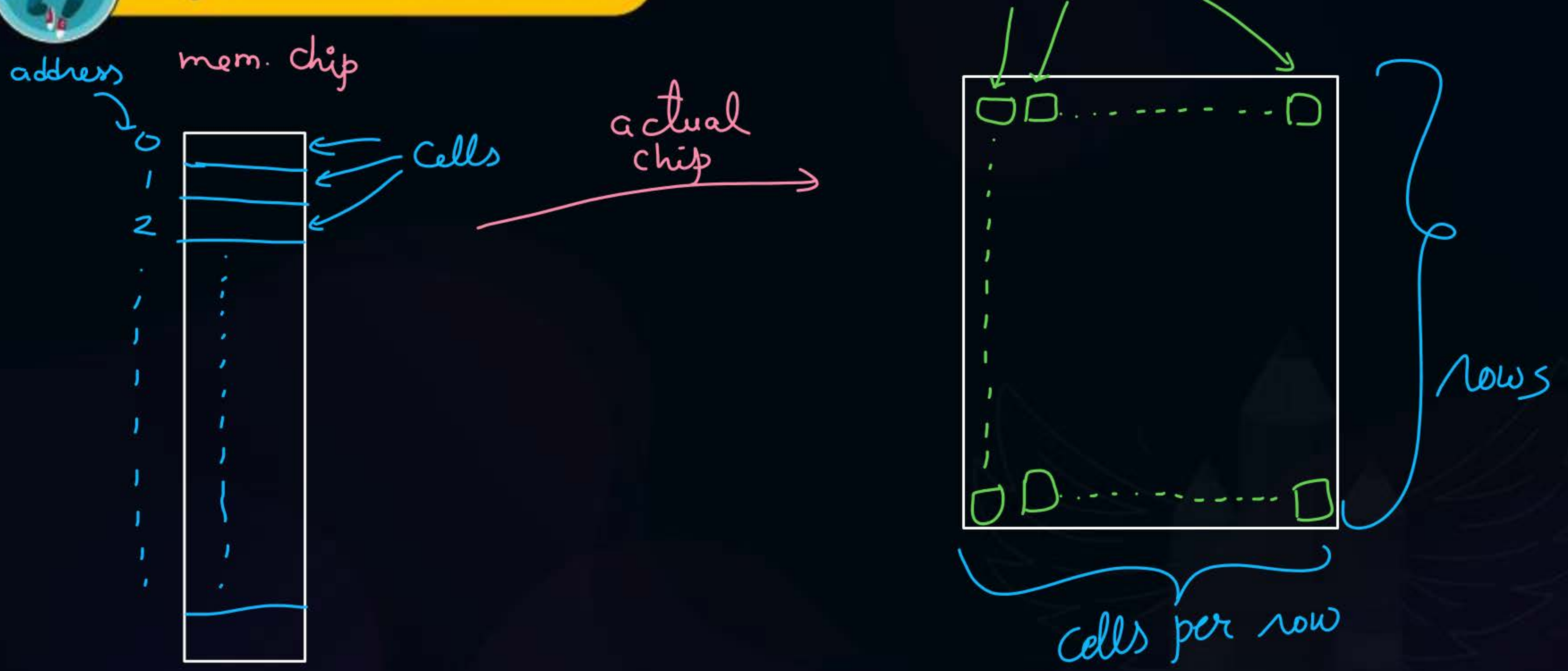
if arrangement ⇒ vertical



arrangement



Topic : DRAM Refresh



chip refresh time = no. of rows of cells in chip * 1 refresh time

n chips mem. system refresh time = 1 chip refresh time

Ques) A chip of size 512×8 bits has 64 rows of cells.

1 refresh takes 5 ns.

Then chip refresh time = 320 ns?

Sol^n

$64 * 5 = 320$ ns

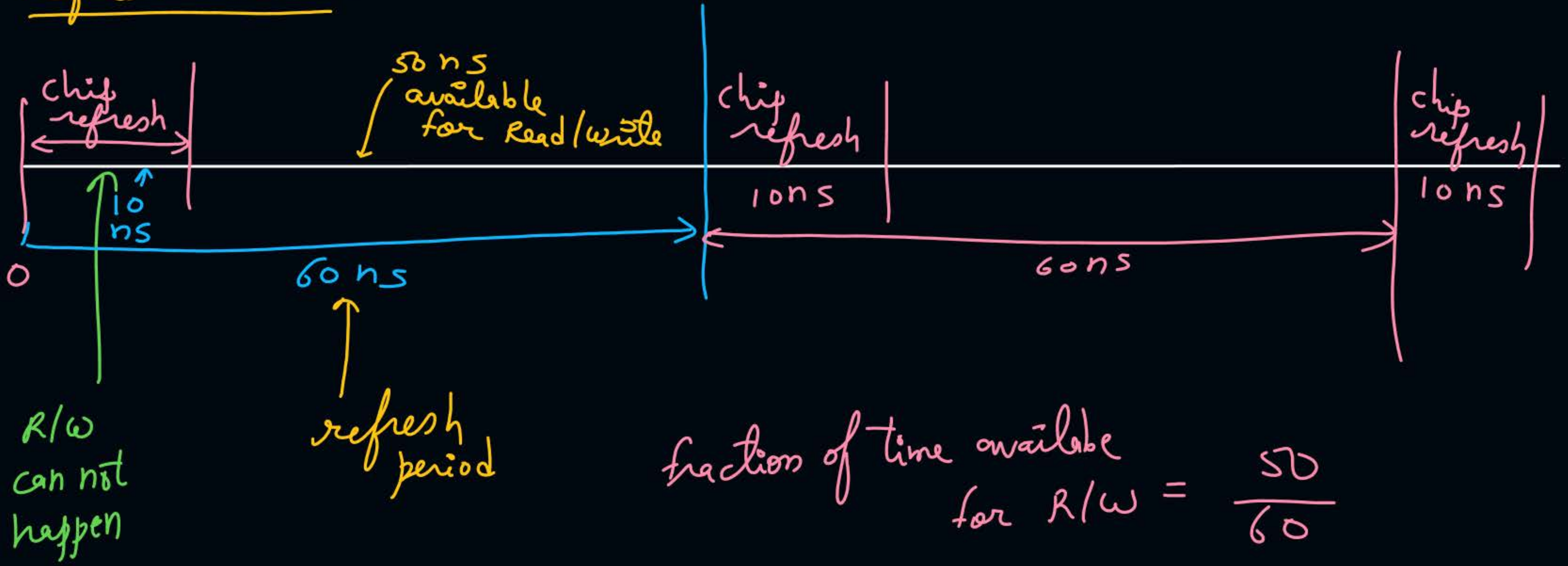
Ques) In above Quest^n, no. of cells per row = 8 ?

Sol^n

$512 = 64 \times x$

$x = \frac{512}{64} = 8$

Refresh Period :-



fraction of time available for R/W $= \frac{50}{60}$

$= \frac{5}{6}$

$= \frac{5}{6} * 100\%$

[NAT]

#Q. Consider a DRAM which can be refreshed in 10ns. The refresh period is 0.05 microseconds. $= 50$ ns

1. % of time taken in refresh? $\frac{10}{50} = 0.2 = 20\%$
2. % of time remaining for read write is? $= \frac{40}{50} = 0.8 = 80\%$

[MCQ]

GATE- 2010

#Q. A main memory unit with a capacity of 4 megabytes is built using 1M × 1–bit DRAM chips. Each DRAM chip has 1K rows of cells with 1K cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is?

A 100 nanoseconds

$= 1K * 100 ns$

B ✓ 100×2^{10} nanoseconds

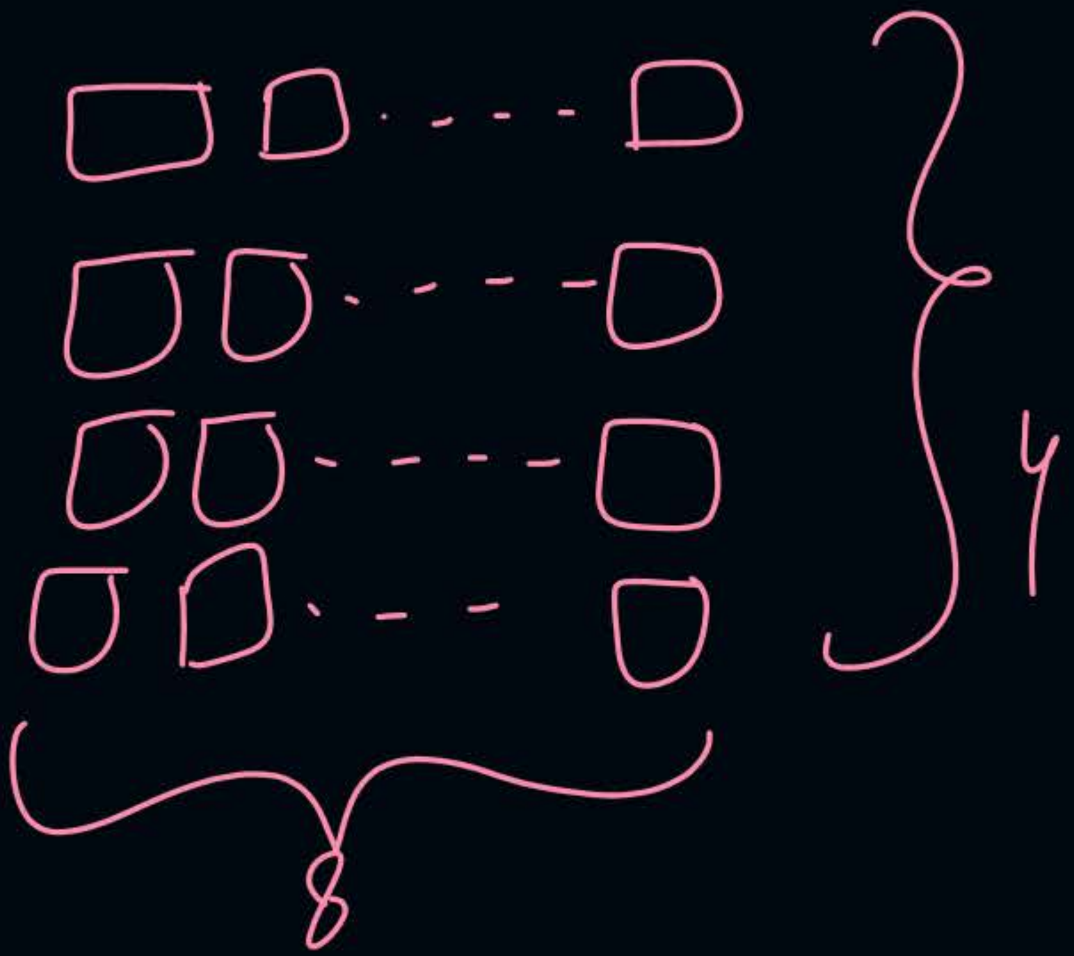
C 100×2^{20} nanoseconds

D 3200×2^{20} nanoseconds

Ques) In prev. Quest^n $\Rightarrow$ how multiple chips arranged?

Sol^n

$$\frac{\overset{4}{\cancel{4M}}\ \overset{8}{\cancel{B}}}{\cancel{1M \times 1\text{ bit}}} = 32 \text{ chips}$$



[NAT]

#Q. A DRAM chip of 128K×8 bits has x rows of cells with y cells in each row? If DRAM takes 20ns for 1 refresh and 10240ns for entire chip refresh then the value of x + y is 768 ?

no. of cells

$$10240^{ns} = x * 20ns$$

$$x = \frac{10240}{20} = 512$$

$$x * y = 128K$$

$$512 * y = 2^{17}$$

$$y = 2^8 = 256$$

$$x + y$$

$$= 512 + 256$$

$$= 768$$

[NAT] GATE- 2018 H.W.

#Q. A 32–bit wide main memory unit with a capacity of 1 GB is built using 256M X 4-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closet integer) of the time available for performing the memory read/write operations in the main memory unit is ______?

2 mins Summary

Topic Multiple Chips in Single Memory System

Topic DRAM Refresh

Happy Learning

THANK - YOU